Advantages

- Pulling power is more
- Can operate vehicles in water, mud, clay and slime areas also
- Sometimes if one wheel fails to function, vehicles can operate using other three wheels

Disadvantages

- Initial cost is high
- Maintenance is high
- Complicated design
- Create more noise
- Vibration is more

1.3.4 Rear engine rear wheel drive

In the rear engine rear wheel drive clutch, gearbox and differential unit are made as a single unit. The engine power is transferred to rear wheel by means of the clutch, gearbox, differential and a small shaft. Propeller Shaft is not used. In this type of drive placing the engine's exhaust system is easy. As the driver is sitting in the front it is difficult to control clutch and gearbox. The drive needs long linkages in this type of drive

Advantages

- Simple in design
- Separate propeller shaft is not needed
- Brake power is high
- No problem to passengers because of exhaust
- No need for lengthy propeller shaft

Disadvantages

- Stability is less
- Long linkages are needed
- Maintenance is high
- Engine cooling is less.

1.4 Air resistance

On the road the blowing wind against the direction of the vehicle and the opposing force due to the vehicle speed oppose the vehicle to move front. This opposing force is called as the air resistance.

To find the air resistance of a vehicle, the following formula can be used.

$$\text{Air resistance } (R_a) = K_a\, A.V^2$$

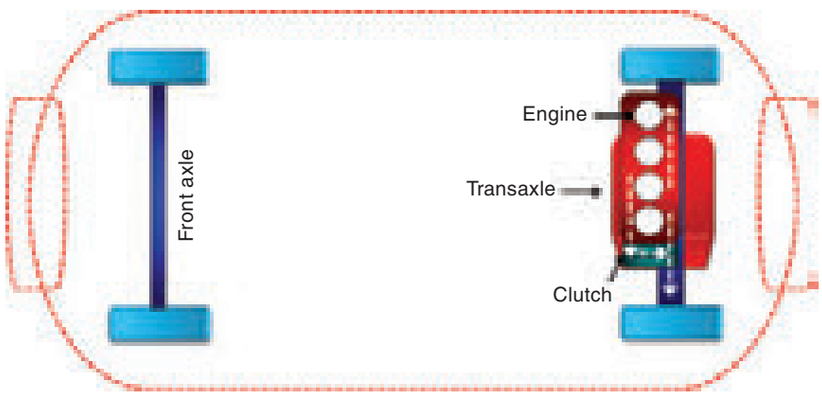


Figure 1.3.4 Rear engine Rear wheel drive

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